

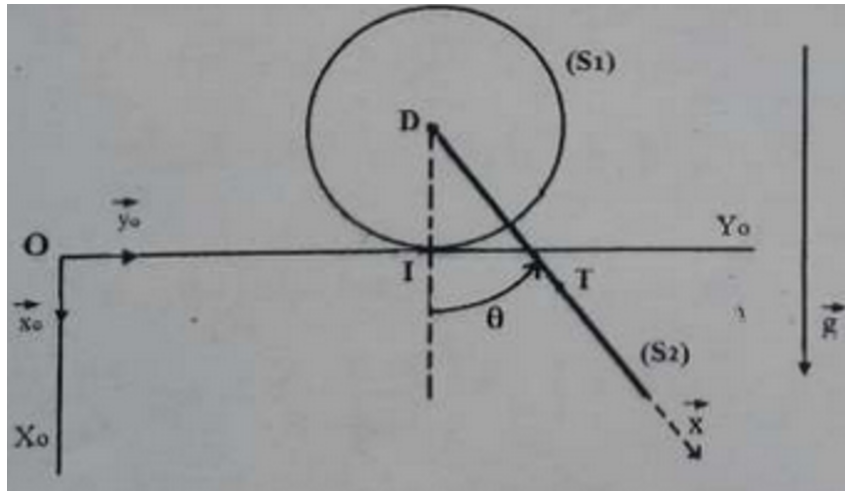
UNIVERSITY NAME

*Rigid Body Mechanics — Make-Up Exam**Year: 2015***Exercise 1:**

Consider a perfect rigid body S composed of a homogeneous disk S_1 of center D , radius a , and mass M , and a homogeneous rod S_2 of center of mass T , length L , and mass m , welded to the disk at point D .

The system is in motion in the vertical plane X_0OY_0 of the reference frame $R_0(O, X_0, Y_0, Z_0)$, with basis $(\bar{x}_0, \bar{y}_0, \bar{z}_0)$, assumed to be fixed and Galilean. The disk S_1 rolls without slipping on the axis OY_0 . I denotes the geometric contact point of S_1 with the axis OY_0 , and y represents the coordinate of point D on the axis OY_0 . The position of the rod relative to the vertical is defined by the angle θ . We denote by $\bar{R}$ the reaction force exerted on the system at point I .

1. (a) State the kinetic energy theorem.
(b) Prove this theorem.
2. Calculate the inertia matrix at point D of the disk S_1 .
3. Calculate the inertia matrix at point T of the rod S_2 and deduce its inertia matrix at point D by applying Huygens' theorem.
4. Calculate the sliding velocity of the solid and deduce the condition for rolling without slipping.
5. Determine, as a function of the angle θ , the absolute velocity of point T , expressed in the basis of R_0 .
6. Calculate the kinetic energy of the system $E_c(S/R_0)$ as a function of θ and $\dot{\theta}$.
7. (a) By applying the kinetic energy theorem to the system in R_0 , determine the first integral of energy.
(b) Deduce the differential equation of motion of the system.
8. Limiting to the case of small angles and keeping only first-order terms in θ and $\dot{\theta}$, specify the nature of the motion and calculate the period of small oscillations T_0 .
9. Retrieve the differential equation of motion of the system by applying the angular momentum theorem at the geometric point I in the reference frame R_0 .



Answer Area

1. (a) **Kinetic energy theorem:** For a rigid body S moving in a Galilean frame R_0 , the time derivative of its kinetic energy equals the power of external forces:

$$\frac{d}{dt} E_c(S/R_0) = P_{\text{ext}} = \sum_i \vec{F}_i \cdot \vec{V}(P_i/R_0)$$

where P_i are the points of application of external forces $\vec{F}_i$.

- (b) **Proof:** Starting from $E_c = \frac{1}{2} \int_S \vec{V}^2(P) dm$, differentiate:

$$\frac{dE_c}{dt} = \int_S \vec{V}(P) \cdot \vec{a}(P) dm$$

Using Newton's second law $\vec{a}(P) dm = d\vec{F}_{\text{ext}}$, the right-hand side becomes the power of external forces.

2. **Inertia matrix of disk S_1 at D :**

For a homogeneous disk of mass M , radius a , thickness negligible, in the plane $(D, \vec{x}_1, \vec{y}_1)$ with $\vec{z}_1$ perpendicular:

$$\mathbf{I}_D(S_1) = \begin{pmatrix} \frac{1}{4}Ma^2 & 0 & 0 \\ 0 & \frac{1}{4}Ma^2 & 0 \\ 0 & 0 & \frac{1}{2}Ma^2 \end{pmatrix}$$

in the basis $(\vec{x}_1, \vec{y}_1, \vec{z}_1)$.

3. **Inertia matrix of rod S_2 :**

Rod length L , mass m , homogeneous, along direction $\vec{u}$:

$$\mathbf{I}_T(S_2) = \frac{1}{12}mL^2 (\mathbf{I} - \vec{u} \otimes \vec{u})$$

At point D , using Huygens' theorem:

$$\mathbf{I}_D(S_2) = \mathbf{I}_T(S_2) + m \left(\|\overrightarrow{DT}\|^2 \mathbf{I} - \overrightarrow{DT} \otimes \overrightarrow{DT} \right)$$

where $\overrightarrow{DT} = \frac{L}{2}\vec{u}$.

4. Sliding velocity and rolling condition:

Let I be geometric contact point between disk and OY_0 . Sliding velocity:

$$\vec{V}_g(I) = \vec{V}(I \in S_1/R_0)$$

Rolling without slipping condition:

$$\vec{V}_g(I) = \vec{0}$$

This gives $\dot{y} = a\dot{\theta}$ if θ is rotation of disk.

5. Velocity of T :

Point T is center of rod, fixed relative to disk at distance a from D along rod direction. If rod is welded at D and makes angle θ with vertical:

$$\overrightarrow{OT} = \overrightarrow{OD} + \overrightarrow{DT}$$

with $\overrightarrow{OD} = y\vec{y}_0$, $\overrightarrow{DT} = L(\sin \theta \vec{x}_0 + \cos \theta \vec{y}_0)$. Velocity:

$$\vec{V}(T/R_0) = \dot{y}\vec{y}_0 + L\dot{\theta}(\cos \theta \vec{x}_0 - \sin \theta \vec{y}_0)$$

6. Kinetic energy of system:

$$E_c = E_c(S_1) + E_c(S_2)$$

Disk: translation of D + rotation about D :

$$E_c(S_1) = \frac{1}{2}M\dot{y}^2 + \frac{1}{2}\left(\frac{1}{2}Ma^2\right)\dot{\theta}^2$$

Rod: translation of T + rotation about T :

$$E_c(S_2) = \frac{1}{2}m\|\vec{V}(T)\|^2 + \frac{1}{2}\left(\frac{1}{12}mL^2\right)\dot{\theta}^2$$

Substitute $\dot{y} = a\dot{\theta}$ from rolling condition.

7. (a) First integral of energy:

External forces: weight (conservative) and reaction at I (no work since rolling without slipping). Therefore total mechanical energy conserved:

$$E_c + E_p = \text{const}$$

with $E_p = Mgy_D + mgy_T$.

(b) Differential equation of motion:

Differentiate energy conservation w.r.t. time:

$$\frac{d}{dt}(E_c + E_p) = 0$$

Yields equation of form $A(\theta)\ddot{\theta} + B(\theta)\dot{\theta}^2 + C(\theta) = 0$.

8. Small oscillations:

For $|\theta| \ll 1$, approximate $\sin \theta \approx \theta$, $\cos \theta \approx 1$, neglect $\dot{\theta}^2$ terms. Equation linearizes to:

$$\ddot{\theta} + \omega_0^2 \theta = 0$$

with

$$\omega_0^2 = \frac{(M+m)ga + mgL/2}{\frac{3}{2}Ma^2 + m(a^2 + \frac{L^2}{3} + aL)}$$

Period:

$$T_0 = \frac{2\pi}{\omega_0}$$

9. Angular momentum theorem at I :

$$\frac{d}{dt} \vec{L}_I(S/R_0) = \vec{M}_I(\text{ext})$$

External forces: weights $M\vec{g}$ at D , $m\vec{g}$ at T , and reaction $\vec{R}$ at I . Moment about I :

$$\vec{M}_I = \overrightarrow{ID} \times M\vec{g} + \overrightarrow{IT} \times m\vec{g}$$

Compute $\vec{L}_I$ using König's theorem or transport formula, equate time derivative to $\vec{M}_I$, recover same equation as in (7b).